University of Ottawa

Department of Mathematics and Statistics

MAT 1302C: Mathematical Methods II

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Midterm 1 – October 11, 2011

Version A

Surname _____ First Name _____

Student # _____ DGD Number _____

Instructions:

(1) You have 80 minutes to complete this exam.
(2) The number of points available for each question is indicated in between parentheses.
(3) All work to be considered for grading should be written in the space provided. The reverse side of pages is for scrap work. If you find that you need extra space in order to answer a particular question, you should continue on the reverse side of the page and indicate this **clearly**. Otherwise, the work written on the reverse side of pages will not be considered for marks.
(4) Write your student number at the top of each page in the space provided.
(5) No notes, books, scrap paper, calculators or other electronic devices are allowed.
(6) You are strongly recommended to write in **pen**, not pencil.
(7) You may use the last page of the exam as scrap paper.

Good luck!

Please do not write in the table below.

Question	1	2	3	4	5	6	Total
Maximum	3	4	2	4	2	5	20
Score							

1. **(3 points)** Find all solutions to the vector equation

$$x_1\begin{bmatrix}1\\3\\-1\end{bmatrix}+x_2\begin{bmatrix}-2\\-6\\2\end{bmatrix}+x_3\begin{bmatrix}2\\1\\1\end{bmatrix}+x_4\begin{bmatrix}5\\0\\4\end{bmatrix}=\begin{bmatrix}0\\1\\-\frac{3}{5}\end{bmatrix}$$

and write the solution in vector parametric form.

Solution: The augmented matrix associated to the system corresponding to the above vector equation is row reduced as follows:

$$\left[\begin{array}{cccc|c}1&-2&2&5&0\\3&-6&1&0&1\\-1&2&1&4&\frac{-3}{5}\end{array}\right]\xrightarrow[R_2\to R_2-3R_1]{R_3\to R_3+R_1}\left[\begin{array}{cccc|c}1&-2&2&5&0\\0&0&-5&-15&1\\0&0&3&9&\frac{-3}{5}\end{array}\right]$$

$$\xrightarrow{R_2\to-\frac{1}{5}R_2}\left[\begin{array}{cccc|c}1&-2&2&5&0\\0&0&1&3&\frac{-1}{5}\\0&0&3&9&\frac{-3}{5}\end{array}\right]\xrightarrow{R_3\to R_3-3R_2}\left[\begin{array}{cccc|c}1&-2&2&5&0\\0&0&1&3&\frac{-1}{5}\\0&0&0&0&0\end{array}\right]$$

$$\xrightarrow{R_1\to R_1-2R_2}\left[\begin{array}{cccc|c}1&-2&0&-1&\frac{2}{5}\\0&0&1&3&-\frac{1}{5}\\0&0&0&0&0\end{array}\right]$$

Thus the system is consistent and the solution is $\begin{cases}x_1=2x_2+x_4+\frac{2}{5}\\x_2:\text{free}\\x_3=-3x_4-\frac{1}{5}\\x_4:\text{free}\end{cases}$

and in vector parametric form:

$$\begin{bmatrix}x_1\\x_2\\x_3\\x_4\end{bmatrix}=x_2\begin{bmatrix}2\\1\\0\\0\end{bmatrix}+x_4\begin{bmatrix}1\\0\\-3\\1\end{bmatrix}+\begin{bmatrix}\frac{2}{5}\\0\\-\frac{1}{5}\\0\end{bmatrix}.$$

2. **(4 points)** Determine if the following linear system is consistent or inconsistent. If it is consistent, then find its general solution.

$$\begin{cases} -3x + 6y - z + w - 7v = 7 \\ x - 2y + 2z + 3w - v = 1 \\ 2x - 4y + 5z + 8w - 4v = 4 \end{cases}$$

Solution: The augmented matrix of the above system of linear equations is

$$A = \left[\begin{array}{rrrrr|r} -3 & 6 & -1 & 1 & -7 & 7 \\ 1 & -2 & 2 & 3 & -1 & 1 \\ 2 & -4 & 5 & 8 & -4 & 4 \end{array}\right]$$

Row reducing one obtains

$$A \xrightarrow{R_1 \leftrightarrow R_2} \left[\begin{array}{rrrrr|r} 1 & -2 & 2 & 3 & -1 & 1 \\ -3 & 6 & -1 & 1 & -7 & 7 \\ 2 & -4 & 5 & 8 & -4 & 4 \end{array}\right] \xrightarrow[R_3 \to R_3 - 2R_1]{R_2 \to R_2 + 3R_1} \left[\begin{array}{rrrrr|r} 1 & -2 & 2 & 3 & -1 & 1 \\ 0 & 0 & 5 & 10 & -10 & 10 \\ 0 & 0 & 1 & 2 & -2 & 2 \end{array}\right]$$

$$\xrightarrow{R_2 \to \frac{1}{5}R_2} \left[\begin{array}{rrrrr|r} 1 & -2 & 2 & 3 & -1 & 1 \\ 0 & 0 & 1 & 2 & -2 & 2 \\ 0 & 0 & 1 & 2 & -2 & 2 \end{array}\right] \xrightarrow{R_3 \to R_3 - R_2} \left[\begin{array}{rrrrr|r} 1 & -2 & 2 & 3 & -1 & 1 \\ 0 & 0 & 1 & 2 & -2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

$$\xrightarrow{R_1 \to R_1 - 2R_2} \left[\begin{array}{rrrrr|r} 1 & -2 & 0 & -1 & 3 & -3 \\ 0 & 0 & 1 & 2 & -2 & 2 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

Hence x, z are basic variables and y, w, v are free, and the general solution is:

$$\begin{cases} x = 2y + w - 3v - 3 \\ y : \text{free} \\ z = -2w + 2v + 2 \\ w : \text{free} \\ v : \text{free} \end{cases}$$

3. **(2 points)** Find the value(s) of h for which the following system has a unique solution:

$$\begin{cases} x + y = 5 \\ 2x + hy = 3 \end{cases}$$

You should justify your answer.

Solution: The augmented matrix of the above system of linear equations is

$$A = \left[\begin{array}{cc|c} 1 & 1 & 5 \\ 2 & h & 3 \end{array}\right]$$

and we can row reduce it as follows:

$$\left[\begin{array}{cc|c} 1 & 1 & 5 \\ 2 & h & 3 \end{array}\right] \xrightarrow{R_2 \to R_2 - 2R_1} \left[\begin{array}{cc|c} 1 & 1 & 5 \\ 0 & h-2 & -7 \end{array}\right]$$

To get a unique solution, one must impose that $h \neq 2$.

4. **(4 points)** Suppose that

$$\mathbf{v}_1 = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \\ -3 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 1 \\ 2 \\ 0 \end{bmatrix}.$$

Is the vector $\mathbf{b} = \begin{bmatrix} 2 \\ 2 \\ 1 \end{bmatrix}$ in $\text{Span}\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$? You should justify your answer.

Solution: Note that the augmented matrix of the corresponding system can be row reduced as follows:

$$\left[\begin{array}{ccc|c} -1 & 1 & 1 & 2 \\ -1 & -1 & 2 & 2 \\ 1 & -3 & 0 & 1 \end{array}\right] \xrightarrow[R_3 \to R_3 + R_1]{R_2 \to R_2 - R_1} \left[\begin{array}{ccc|c} -1 & 1 & 1 & 2 \\ 0 & -2 & 1 & 0 \\ 0 & -2 & 1 & 3 \end{array}\right]$$

$$\xrightarrow{R_3 \to R_3 - R_2} \left[\begin{array}{ccc|c} -1 & 1 & 1 & 2 \\ 0 & -2 & 1 & 0 \\ 0 & 0 & 0 & 3 \end{array}\right] \xrightarrow[R_3 \to \frac{1}{3}R_3]{R_2 \to -\frac{1}{2}R_2} \left[\begin{array}{ccc|c} -1 & 1 & 1 & 2 \\ 0 & 1 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 1 \end{array}\right]$$

$$\xrightarrow[R_1 \to R_1 - R_3]{R_1 \to R_1 - R_2} \left[\begin{array}{ccc|c} -1 & 0 & \frac{3}{2} & 0 \\ 0 & 1 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 1 \end{array}\right] \xrightarrow{R_1 \to -R_1} \left[\begin{array}{ccc|c} 1 & 0 & -\frac{3}{2} & 0 \\ 0 & 1 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 1 \end{array}\right]$$

Therefore the rightmost column is a pivot column, and it follows that the answer is negative.

5. **(2 points)** For what value(s) of h are the vectors

$$\mathbf{v}_1 = \begin{bmatrix} -1 \\ 2 \\ 4 \end{bmatrix}, \mathbf{v}_2 = \begin{bmatrix} 3 \\ -7 \\ -6 \end{bmatrix}, \mathbf{v}_3 = \begin{bmatrix} 2 \\ 1 \\ h \end{bmatrix}$$

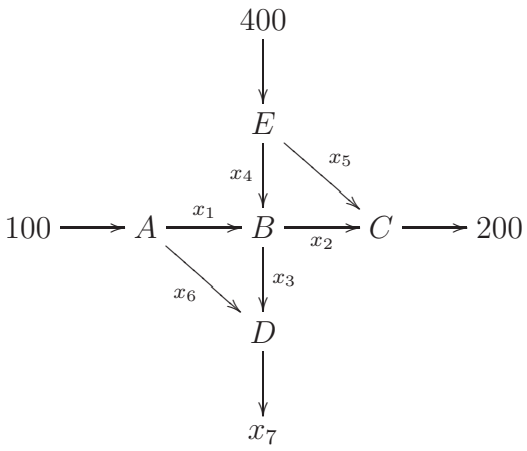
linearly independent? You should justify your answer.

Solution: Note that

$$\begin{bmatrix} -1 & 3 & 2 \\ 2 & -7 & 1 \\ 4 & -6 & h \end{bmatrix} \xrightarrow[R_3 \to R_3 + 4R_1]{R_2 \to R_2 + 2R_1} \begin{bmatrix} -1 & 3 & 2 \\ 0 & -1 & 5 \\ 0 & 6 & h+8 \end{bmatrix} \xrightarrow{R_3 \to R_3 + 6R_2} \begin{bmatrix} -1 & 3 & 2 \\ 0 & -1 & 5 \\ 0 & 0 & h+38 \end{bmatrix}$$

If $h \neq -38$ then every column is a pivot column and vice versa, in which case the vectors are linearly independent.

6. **(5 points)** Consider the traffic flow described by the following diagram. The letters A through E label intersections. The arrows indicate the direction of flow (all roads are one-way) and their labels indicate flow in cars per minute.



(a) Write down a linear system describing the traffic flow, i.e., all constraints on the variables $x_i, i = 1, \ldots, 7$. (Do not solve the linear system at this stage.)

Solution:
Note that x_i are positive integers, $x_i \geq 0$ for all $i = 1, 2, \ldots, 7$. Moreover:
$100 = x_1 + x_6$
$x_1 + x_4 = x_2 + x_3$
$x_5 + x_2 = 200$
$x_3 + x_6 = x_7$
$400 = x_4 + x_5$
$100 + 400 = 200 + x_7$

(b) The reduced row echelon form of the augmented matrix corresponding to the linear system in part (a) is:

$$\left[\begin{array}{ccccccc|c} 1 & 0 & 0 & 0 & 0 & 1 & 0 & 100 \\ 0 & 1 & 0 & 0 & 1 & 0 & 0 & 200 \\ 0 & 0 & 1 & 0 & 0 & 1 & 0 & 300 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 400 \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 300 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

Find the general flow pattern (i.e., the general solution of the linear system).

Solution: x_1, x_2, x_3, x_4, x_7 are basic and x_5, x_6 are free, and the general solution is:

$$\begin{cases} x_1 = 100 - x_6 \\ x_2 = 200 - x_5 \\ x_3 = 300 - x_6 \\ x_4 = 400 - x_5 \\ x_5 : \text{free} \\ x_6 : \text{free} \\ x_7 = 300 \end{cases}$$

(c) Suppose that due to road work, the flow along AD is limited to a maximum of 50 cars per minute. What is the minimum possible flow along BD?

Solution: Since $x_6 \leq 50$ and $x_3 = 300 - x_6$ we obtain $x_3 \geq 300 - 50 = 250$

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